Charles Shang

Principal ML Engineer - Generative Video, World Models, and 3D Generation

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PROFILE

Hands-on ML research engineer with 10+ years in computer vision and machine learning, specializing in 3D generation, generative video, and world models. Work directly across the full model development lifecycle: representation and tokenizer design, data pipelines, distributed training, evaluation, and production inference. Co-developed Roblox Cube and took Cube MeshGPT from research experiments to production for controllable 3D asset generation. Scaled video model training to hundreds of GPUs and shipped real-time avatar models on client devices and in the cloud.

EXPERIENCE

Roblox California
Principal Software Engineer *2021-Present*

VideoGen Action-conditioned real-time video and video-to-video generation

- Built the team's first video data pipeline: ingestion, filtering, deduplication, captioning, latent extraction, and automated quality scoring.
- Scaled training to hundreds of GPUs on a shared PyTorch stack with FSDP, tensor parallelism, and sequence parallelism.
- Trained action-conditioned real-time video and video-to-video models for controllable generation and long-horizon rollouts.
- Designed bucketed eval sets and model-based scoring that isolate temporal and long-horizon failure modes, and drove the next round of model and data work off those results.

MeshGen (Roblox Cube) Discrete representations and autoregressive mesh generation

- Advanced 3D representation learning by co-designing mesh tokenization, the training recipe, and the data pipeline, closing most of the quality gap between discrete VQ and continuous KL-VAE representations.
- Took Cube MeshGPT from research experiments to production for text- and bounding-box-conditioned 3D asset generation, spanning model training, evaluation, and inference.

Production ML Real-time avatars and safety

- Shipped models for voice moderation, audio-driven facial animation, lip sync, face tracking, and personalized avatars, running on client devices and in the cloud.

Loomai - *Deep Learning Consultant, Part-time* *Remote / Beijing, 2019-2021*
Built compact on-device models for audio-driven facial animation, activity detection, and speaker verification.

Institute of Automation, Chinese Academy of Sciences - *Postdoctoral Researcher* *Beijing, 2019-2020*
Built reinforcement-learning systems for games; mentored work on detection, tracking, and pose estimation.

Earlier research internships. Microsoft Research Asia (2017), representation learning; Intel (2015-2016), visual recognition.

TECHNICAL STRENGTHS

Modeling: 3D representation learning, mesh tokenization, autoregressive mesh generation (MeshGPT), VQ and KL-VAE; action-conditioned video, video-to-video, world models, diffusion and flow matching

Training: PyTorch, FSDP, tensor and sequence parallelism, training on hundreds of GPUs, distributed evaluation, inference optimization

Systems: Python, C/C++, CUDA, Linux, Ray, data pipelines, client and cloud inference

SELECTED RESEARCH AND OPEN SOURCE

2025 - Cube: A Roblox View of 3D Intelligence. Foundation AI Team, Charles Shang, et al. arXiv:2503.15475.

2025 - Efficient Autoregressive Shape Generation via Octree-Based Adaptive Tokenization. Deng, Liu, Zhu, Sun, Charles Shang, et al. arXiv:2504.02817.

2024 - Diffusion Synthesizer for Efficient Multilingual Speech-to-Speech Translation. Hirschkind et al., Charles Shang, et al. Interspeech.

Roblox Cube. Co-developed and open-sourced a foundation model for 3D shape generation and understanding; 1.2K+ stars.

ML research implementations. Mask R-CNN, Deformable ConvNets v2, and Faster R-CNN, reimplemented and open-sourced; 5K+ stars and 1.8K+ forks.

EDUCATION

Tsinghua University - *Ph.D., Computer Vision and Machine Learning; advisor: Prof. Haizhou Ai* *Beijing, 2013-2019*

Northwestern Polytechnical University - *B.E., Computer Science; National Scholarship* *Xi'an, 2008-2013*